

5 (a) Find the volume scale factor of the transformation with associated matrix $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & -1 \\ -1 & 0 & 2 \end{pmatrix}$. [2]

(b) The transformations S and T of the plane have associated 2×2 matrices **P** and **Q** respectively.

(i) Write down an expression for the associated matrix of the combined transformation S followed by T. [1]

The determinant of **P** is 3 and $\mathbf{Q} = \begin{pmatrix} k & 3 \\ -1 & 2 \end{pmatrix}$, where k is a constant.

(ii) Given that this combined transformation preserves both orientation and area, determine the value of k . [3]